

NO.

Date.

1. 1 { 1 }

2 { 1 2 }

3 { 1 2 3 }

4 { 1 2 3 4 }

5 { 1 2 5 }

6 { 1 2 5 6 }

2. back edge: $4 \rightarrow 3$
 $5 \rightarrow 2$

3. $PRE(4) = \{5\}$

End

∴ natural loop:
 $\{5, 2, 3, 4\}$

3. natural loop:

for $4 \rightarrow 3$
initial:
stack: {4}
loop: {3, 4}

1. $PRE(4) = 3$

insert(3)

loop: {3, 4}

end ∴ natural loop: {3, 4}

for $5 \rightarrow 2$

initial stack: {5}

loop {5, 2}

1. $PRE(5) = \{3, 2\}$

stack: {3}

loop {5, 2, 3}

2. $PRE(3) = \{4, 2\}$

stack: {4}

loop {5, 2, 3, 4}